



Symmetric Binary Tree Based Co-occurrence Texture Pattern Mining for Fine-grained Plant Leaf Image Retrieval

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ABSTRACT

Leaf image patterns have been actively researched for plant species recognition. However, as a very challenging fine-grained pattern identification issue, cultivar recognition in which the leaf image patterns usually have very subtle difference among cultivars has not yet received considerable attention in computer vision and pattern recognition community. In this paper, a novel symmetric geometric configuration, named Symmetric Binary Tree (SBT) which has multiple symmetric branch pairs and can change in size, is designed to mine the multiple scale co-occurrence texture patterns. The resulting SBT descriptors encode both shape and texture features which make them more informative than the existing individual descriptors and co-occurrence features. A novel feature fusion scheme, named K-NN Based Handcrafted and Deep Features Fusion (KNN-HDFF) that encodes the neighbouring information of distance measure, is proposed for further boosting the retrieval performance. Extensive experiments conducted on the challenging soybean cultivar leaf image dataset and peanut cultivar leaf image dataset consistently indicate the superiority of the proposed method over the state-of-the-art methods on fine-grained leaf image retrieval. We also conduct extensive experiments of feature fusions using the proposed KNN-HDFF on the benchmark datasets and the experimental results prove its potential for improving the performance of cultivar identification which also indicates that fusing handcrafted and deep features may be the direction to address the challenging fine-grained image recognition problem.

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1. Introduction

Leaf plays an important role for botanists to identify species and extensive studies have been done for plant species recognition by analyzing leaf image patterns [34]. Although most of them made great progress on using leaf patterns for plant species recognition, their performance are usually poor when applied to the fine-grained leaf image recognition problem like cultivar identification in which all the leaves share the same species making their image patterns have high similarity among different cultivars. Fig. 1 presents several example leaf images taken from plant species leaf image dataset [1] (see the first row), the soybean cultivar leaf dataset [31] (see the second row) and our collected peanut cultivar leaf dataset (see the third row). It can be observed that both the soybean cultivar leaves and the peanut cultivar leaves have higher inter-class similarity which make the cultivar recognition problem much more challenging than general species recognition.

As a very fine-grained pattern recognition, cultivar recognition contributes to the modern agriculture in many aspects. For instance, as a globally important crop, soybean has many cultivars (varieties) and experts in agriculture always try to develop new cultivars with high productivity and profitability to the soybean industry. Under low-input and organic farming conditions, cultivar choice and breeding are regarded as major tools for improving crop performance [8]. The identification of soybean cultivars accordingly plays a vital role in the evaluation, selection, and production of soybean varieties [15]. In additional, soybean cultivar recognition also contributes to the study of plant phenotype [17].

However, analyzing leaf image patterns for cultivar identification has not yet received considerable attentions in computer vision and pattern recognition community. The reasons behind it are: (1) The very low classification accuracies of the existing algorithms on cultivar identification make researchers have a concern about whether leaf image patterns can provide sufficient information for distinguishing cultivars. (2) Different from the leaf species recognition which can be done by general public, cultivars are usu-

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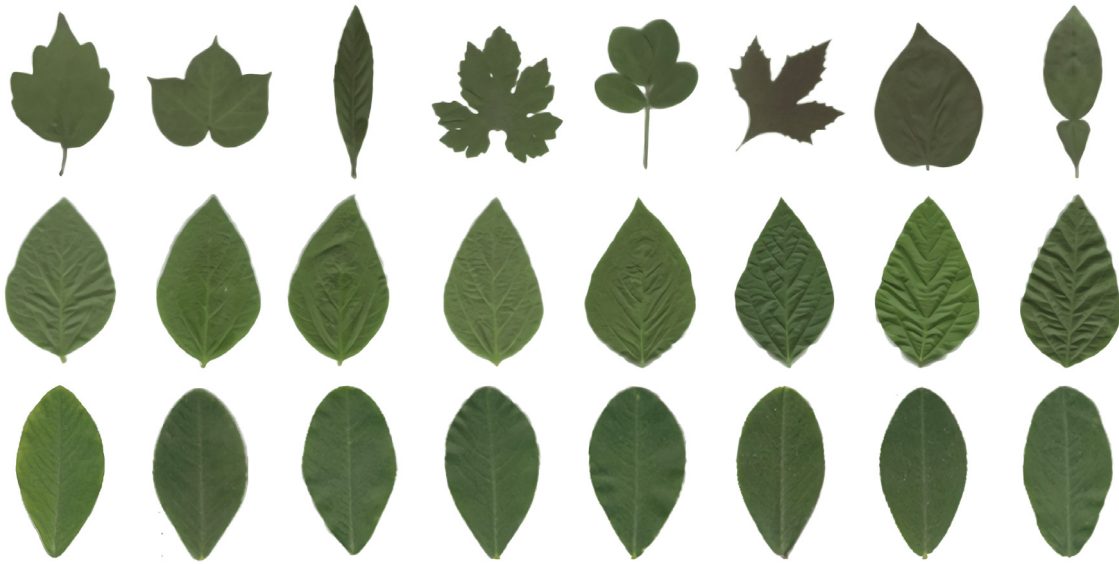


Fig. 1. Comparison of inter-class variation between cultivar leaves and species leaves. From the first row to the third row: example leaves from the plant species leaf image dataset [1], example leaves from the soybean cultivar leaf dataset [31], and our collected peanut cultivar leaves.

ally recognized only by experts with the help of biotechnologies like gene sequencing which makes the collection and annotation of fine-grained leaf images very expensive.

In this paper, we focus on fine-grained leaf image retrieval and try to address the challenging issue of cultivar identification. The early version of this study was initially reported in [35]. Here we present the complete work with new formulation and extensive investigations. The contributions of this work are highlighted as: (1) We propose a novel symmetric geometric configuration, named Symmetric Binary Tree, to mine the co-occurrence texture patterns for constructing a powerful feature representation. (2) We propose a novel feature fusion method, named K-NN Based Handcrafted and Deep Features Fusion (KNN-HDFF), for boosting the accuracy on fine-grained leaf image retrieval. (3) We conduct extensive experiments on fine-grained leaf image datasets including soybean cultivar dataset and peanut cultivar dataset and the proposed method achieves better performance over the state-of-the-art methods including handcrafted descriptors and deep features. The experimental results also indicate the potential of leaf image patterns for fine-grained leaf image retrieval.

The remainder of this paper is organized as follows. The representative studies related to this work are reviewed in Section 2. The proposed method is detailed in Section 3. The experimental results and discussions are presented in Section 4. The conclusions are made in Section 5.

2. Related Work

In the past decades, many methods have been devoted to the identification of plants in species level, or more fine-grained cultivar level through analysing leaf image patterns and achieved great progress. In terms of the scheme of leaf visual features extraction, the existing methods can be divided into two groups: handcrafted descriptors and deep features. The former is manually designed based on the expert knowledge. While the later directly learn visual feature representations from the raw image pixel using deep neural networks.

Plants usually have significant leaf shape variations across different species which play a dominant role in species recognition [19]. Shape contexts [4], Inner-Distance Shape Contexts [5], Hierarchical String Cuts (HSC) [1], Multiscale Distance Matrix [6], Multiscale Arch Height [7] are popular descriptors that extract shape

features from object contour and get promising accuracies for plant species identification. Recently, a novel shape descriptor, Histogram of Gaussian convolution vectors (HoGCV) [3] has been designed to extract boundary features not only for plant species recognition, but also for the more challenging problem of plant cultivar identification. In addition to characterizing leaf shape from the boundary, there are also some methods capturing the inner region features of leaf shapes, such as image moments [20,21], 2D Fourier power spectrums [22], and chord bunch walks [23] which show robust yet accurate performance on leaf image identification.

Beside shape features, texture patterns have also received considerable attention for designing leaf image descriptors in handcrafted features community. Chaki et al. [24] modelled leaf texture using Gabor filter and gray level co-occurrence matrix (GLCM) which are further combined with shape features for leaf image classification. Tang et al. [9] combined a non-overlap window local binary pattern (LBP) and Gray Level Co-Occurrence Matrix (GLCM) for green tea leaves classification. The advantage of this method is its much lower computational load than that of classic LBP [2] and GLCM which make it meet automatic tea production line requirements. Naresh et al. [10] modified the classic LBP by considering the proximity of the neighbouring pixels to the mean and standard deviation of the whole neighbourhood instead of hard threshold used in the basic LBP for the classification of medicinal plants. More recently, Yang [27] combined the leaf texture feature, local binary pattern histogram Fourier (LBP-HF), with the leaf shape feature, multiscale triangle descriptor, for plant species recognition and achieved excellent performance on three public leaf image datasets.

Spatial co-occurrence among features are known for having the potential to increase the discriminative power of features. To address the issue of transformation sensitivity in spatial co-occurrence features, Qi et al. [16] developed a novel Pairwise Rotation Invariant Co-occurrence Local Binary Pattern (PRICoLBP) feature which can not only capture the spatial context co-occurrence information effectively, but also be invariant to rotation. This method reported classification accuracy of 99.38% on the Swedish leaf dataset.

The veins are the vascular tissue of the leaf and appear in an interconnecting network which has been regarded as an important morphological structure by botanist to provide clues for plant classification. Charters [25] proposed a new descriptor exploiting

the edge patterns among neighbouring regions to characterise the overall venation structure. They reported a classification accuracy improvement of 6% to the SURF local descriptor using this method on the Swedish leaf image dataset. Larese et al. [26] extracted a lot of morphological features from the segmented venation and shown its superior performance on legumes species classification and reported an average accuracy of 60.20% on distinguishing three diverse soybean cultivars.

The aforementioned methods classify plants only considering single leaf pattern which may not be informative enough to handle the challenging recognition tasks of cultivar level. More recently, Wang et al. [31] made the first attempt to study the complementarity of leaf image patterns from different parts of plants. In their work, a novel multiscale sliding chord matching (MSCM) approach is first designed for depicting shape and appearance features of single leaf pattern. The MSCM image descriptors of the leaves from the lower, middle and upper parts of plants are then combined to form a joint feature representation for enhancing the discriminative power of leaf image descriptors. They built a soybean leaf dataset, named SoyCultivar200, consisting of 6000 samples of 200 soybean cultivars that are collected from the lower, middle and upper parts of soybean plants, for the performance evaluation of cultivar recognition. They reported the best classification accuracies of 34.70% for using single leaf patterns, and the classification accuracy of 72.40% for using joint leaf patterns.

Recently, there are many efforts aiming to mine deep features using deep convolutional neural network (CNN). Lee et al. [11] used CNN to learn useful leaf features directly from the raw representation of input data and gained intuition of the chosen features from Deconvolutional Network (DN). For providing an efficient automatic plant species classification, Tan et al. [12] proposed a novel CNN model to combine multiple pre-trained network models such as pre-trained AlexNet and fine-tuned AlexNet for leaf feature extraction. Grinblat et al. [13] focused on using CNN to extract leaf vein features for classifying three different legume species: white bean, red bean, and soybean and attained a mean classification accuracy of 96.90% in their experimental study. More recently, Tavakoli et al. [38] used CNN to classify 12 different cultivars of common beans that belong to three various species. Instead of using the standard softmax loss function, they used the advance loss functions, like Additive Angular Margin Loss and Large Margin Cosine Loss being adopted for improving the discriminative power.

3. The Proposed Method

Given a leaf image, its gray-scale version can be denoted as a 2D function $g(x, y) : \mathcal{R} \times \mathcal{R} \rightarrow [0, 1]$ that describes the intensity of a pixel at Cartesian coordinates x and y in the image plane which has been normalized with respect to its largest gray level. The outer contour Ω of the leaf can be represented as a curve defined by an arch-length parameterization form [14] $z(\tau) = (\bar{x}(\tau), \bar{y}(\tau)) \in \Omega$ with $\tau \in [0, 1]$ being the normalized perimeter length of the contour from the starting point of the contour in anticlockwise direction.

3.1. Symmetric Binary Tree (SBT)

Given a point $z(\tau) = (\bar{x}(\tau), \bar{y}(\tau)) \in \Omega$ and a positive number $\rho \in (0, R(\tau))$ with

$$R(\tau) = \max_{t \in [0, 1]} \left[(\bar{x}(t) - \bar{x}(\tau))^2 + (\bar{y}(t) - \bar{y}(\tau))^2 \right]^{\frac{1}{2}} \quad (1)$$

being its largest distance to the other contour points. Take the point $z(\tau)$ as a center and the number ρ as a radius to draw a circle C in the image plane. Since $\rho < R(\tau)$ and the contour Ω is closed, the circle intersects with the contour Ω by at least two

points. We select two intersections $z(\tau_r)$ and $z(\tau_l)$ that are closest to the point $z(\tau)$ along the contour in anticlockwise and clockwise, respectively. Obviously, the point pairs, $z(\tau_r)$ and $z(\tau_l)$, are uniquely determined by the parameters τ and ρ .

Let $\alpha_{\tau, \rho}$ be the angle formed by rotating the side $\overline{z(\tau)z(\tau_r)}$ about the point $z(\tau)$ to the side $\overline{z(\tau)z(\tau_l)}$ in anticlockwise direction. We emanate a line segment $T_{\tau, \rho}$ from the point $z(\tau)$ that has the length ρ to equally partition the angle $\alpha_{\tau, \rho}$. Then we treat it as the tree trunk, the point $z(\tau)$ as the tree root, the side pair $\overline{z(\tau)z(\tau_r)}$ and $\overline{z(\tau)z(\tau_l)}$ as the branch pair from the root of the trunk $T_{\tau, \rho}$ denoted by $B_{\tau, \rho}^{r, 0}$ and $B_{\tau, \rho}^{l, 0}$, respectively, and the angle $\alpha_{\tau, \rho}$ as the spreading angle to form a binary tree-like geometric structure. We name it Symmetric Binary Tree (SBT). Since it only has one branch pair, $B_{\tau, \rho}^{r, 0}$ and $B_{\tau, \rho}^{l, 0}$, that spread from the root $z(\tau)$, we also term it 1-order SBT. To construct a M -order SBT, we continuously move the shared point $z(\tau)$ of the 0-order branch pair $B_{\tau, \rho}^{r, 0}$ and $B_{\tau, \rho}^{l, 0}$ along the tree trunk $T_{\tau, \rho}$ by $M - 1$ times with the step length being equal to ρ/M , and remove the parts of branch pair that are outside of the circle C which leads to multiple orders of branch pair, $B_{\tau, \rho}^{r, m}$ and $B_{\tau, \rho}^{l, m}$ available, where m is the index of the order of the branch pair. Then a M -order SBT is mathematically expressed as

$$SBT_{\tau, \rho}^M = \left\{ z(\tau), T_{\tau, \rho}, \alpha_{\tau, \rho}, (B_{\tau, \rho}^{r, m}, B_{\tau, \rho}^{l, m})_{m=0, \dots, M-1} \right\}, \quad (2)$$

whose elements corresponds to the root, trunk, spreading angle and branch pairs of M orders, respectively. In Fig. 2, we present an example to show a 4-order SBT rooted at a contour point $z(\tau)$.

From the process of generating SBT, we can see that it is uniquely controlled by three parameters, the number of orders M , the index τ of the contour point $z(\tau)$ and the radius ρ of the circle C . Varying the value of the parameter M will change the density of the branch pairs of the SBT. While the parameter ρ plays a role of scale used to control the size of the SBT. Since $\rho \in (0, R(\tau))$ and $R(\tau)$ varies with the parameter τ (see Eq. (1)), instead of directly using the absolute scale ρ , we introduce a relative scale parameter $s = 1, \dots, S$, which has the following relation with the original parameter ρ :

$$\rho = R(\tau)/2^s, \quad (3)$$

where S is the number of the scale levels. For example, setting the relative scale parameter $s = 1$ means the parameter $\rho = R(\tau)/2$ being used to yield a SBT at the contour point $z(\tau)$. Then we replace the parameter ρ with s and rewrite $SBT_{\tau, \rho}^M$ as $SBT_{\tau, s}^M$ in Eq. (2). We present an example in Fig. 3 to show $SBT_{\tau, s}^M$ at different scale s . As for the parameter τ , it is associated with the position of SBT along the contour Ω . In Fig. 4, we present an example to show the Symmetric Binary Trees (SBTs) of the same scale level ($s=2$) rooted at different positions of the contour Ω .

3.2. Mining Co-occurrence Texture Pattern Using SBT

The proposed SBT is a symmetric geometric configuration that is applicable for mining co-occurrence texture patterns from the textured object images. In this section, we first use it to construct gray-level co-occurrence matrices (GLCMs) and then calculated the statistical features of GLCMs as final feature representations.

For each contour point $z(\tau)$ and a given scale level $s = 1, \dots, S$, we construct a M -order SBT: $SBT_{\tau, s}^M = \{z(\tau), T_{\tau, s}, \alpha_{\tau, s}, (B_{\tau, s}^{r, m}, B_{\tau, s}^{l, m})_{m=0, \dots, M-1}\}$. For each branch pair $(B_{\tau, s}^{r, m}, B_{\tau, s}^{l, m})$, $m = 0, \dots, M - 1$, we can generate a set of pixel point pairs defined as

$$BP = \left\{ (b^r, b^l) \mid b^r \in B_{\tau, s}^{r, m}, b^l \in B_{\tau, s}^{l, m}, \|b^r - z(\tau)\|_2 = \|b^l - z(\tau)\|_2 \right\}, \quad (4)$$

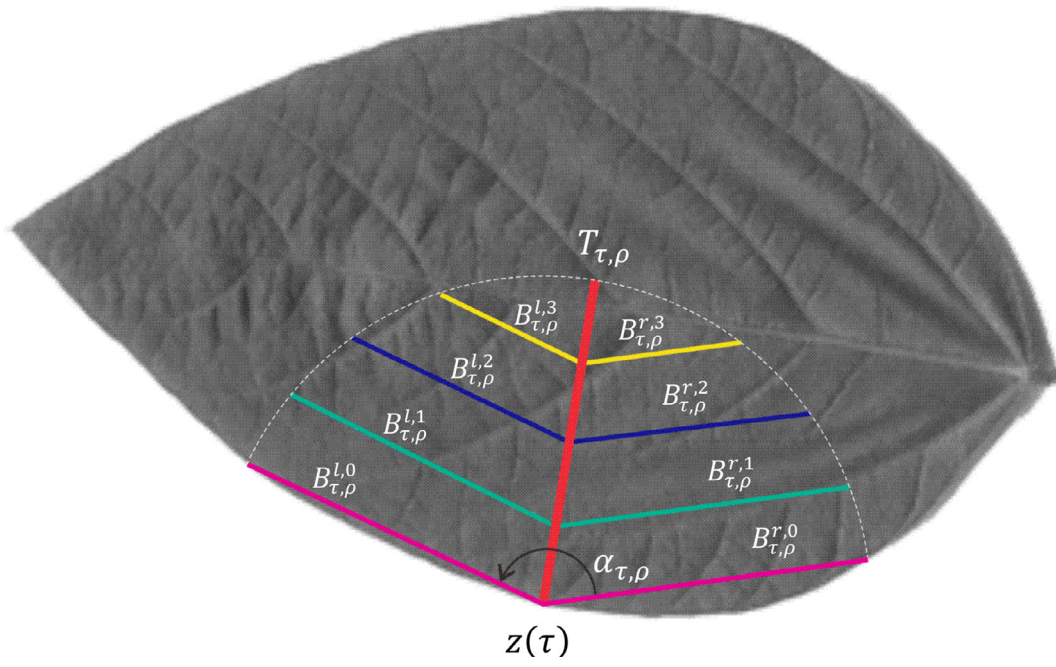


Fig. 2. An example of 4-order Symmetric Binary Tree (SBT) rooted at the contour point $z(\tau)$.

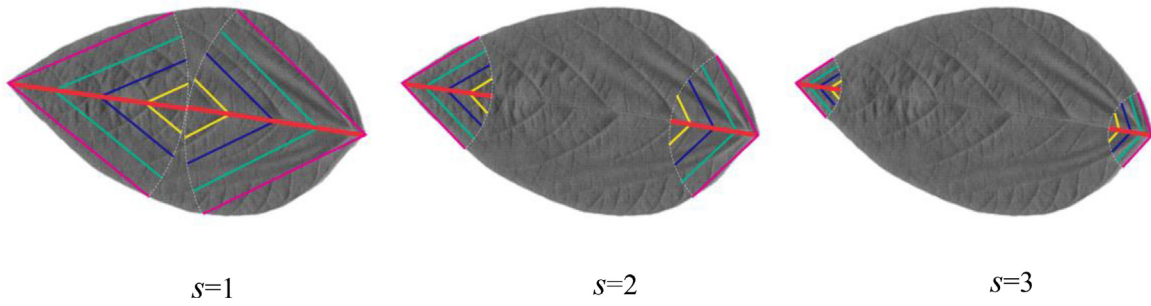


Fig. 3. An example to show Symmetric Binary Trees (SBTs) at different scale levels.

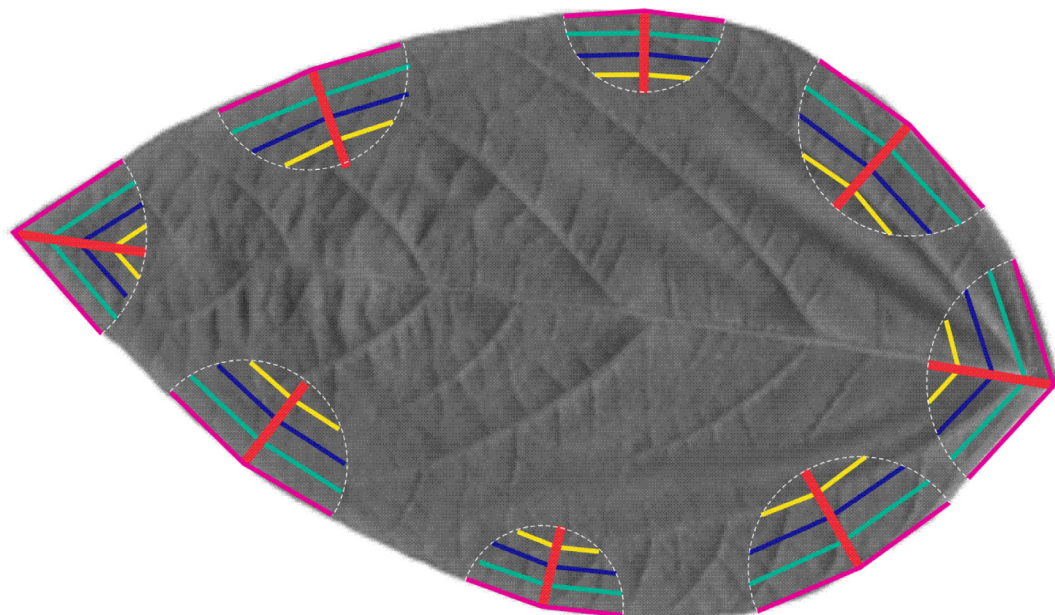


Fig. 4. An example to show the Symmetric Binary Trees (SBTs) of the same scale level ($s=2$) rooted at different positions of the contour Ω .

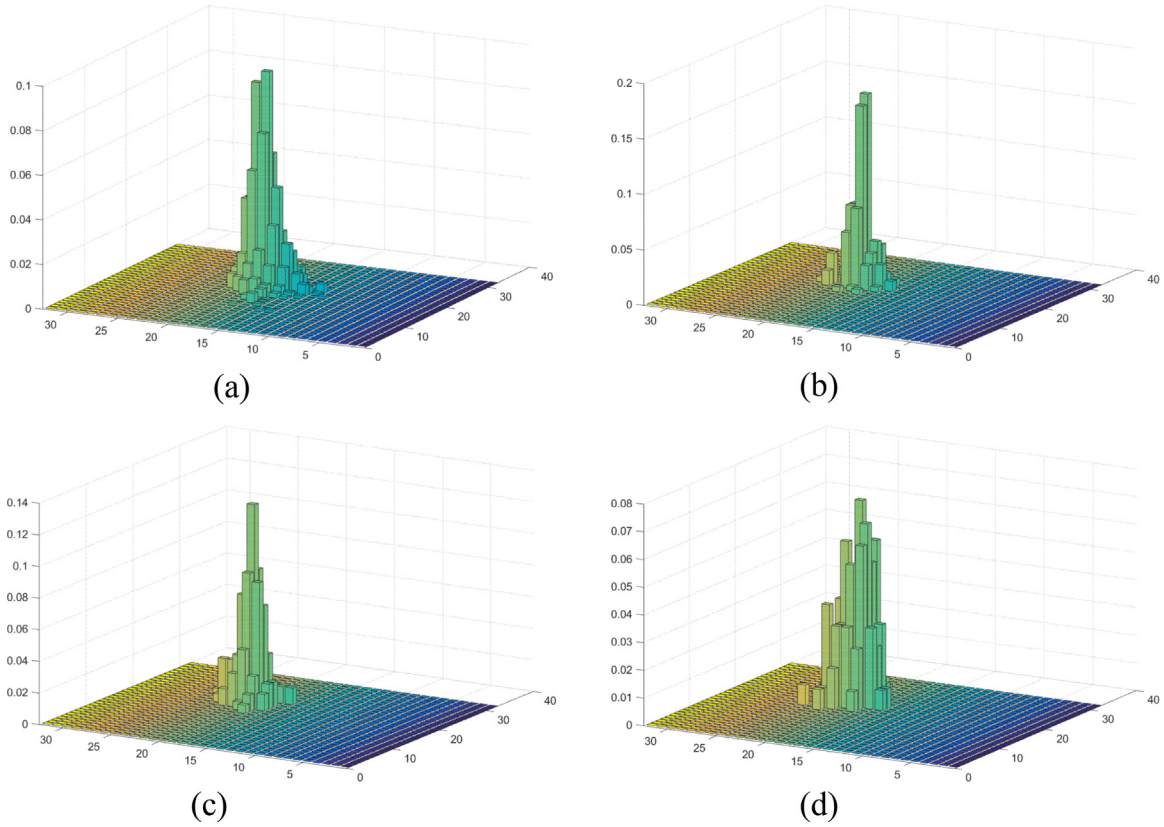


Fig. 5. A visualization of the gray-level co-occurrence matrices, (a) $GLCM_{\tau,1}^0$, (b) $GLCM_{\tau,1}^1$, (c) $GLCM_{\tau,1}^2$ and (d) $GLCM_{\tau,1}^3$ derived from the 4-order Symmetric Binary Tree (SBT) shown in Fig. 2, where the size of the GLCM is set to 32×32 .

where $\|b^r - z(\tau)\|_2$ denotes the Euclidean distance between the points b^r and $z(\tau)$. From the definition of the set BP, we can see that for each pixel point pair (b^r, b^l) in BP, the pixels b^r and b^l are symmetrical about the tree trunk $T_{\tau,s}$. We uniformly divide the grayscale range $[0, 1]$ into Q intervals, $G_1, \dots, G_Q$. Then a gray-level co-occurrence matrix $GLCM_{\tau,s}^m$ of size $Q \times Q$ constructed from the branch pair $(B_{\tau,s}^{r,m}, B_{\tau,s}^{l,m})$ is defined as

$$GLCM_{\tau,s}^m(u, v) = \frac{\#\{(b^r, b^l) | (b^r, b^l) \in BP, g(b^r) \in G_u, g(b^l) \in G_v\}}{\#BP}, \quad (5)$$

where $\#\{\cdot\}$ denotes the cardinality of the set. The value of $GLCM_{\tau,s}^m(u, v)$ stands for the probability of occurrence of gray-level pair (u, v) in the branch pair. In Fig. 5, we visualize the $GLCM_{\tau,1}^0$, $GLCM_{\tau,1}^1$, $GLCM_{\tau,1}^2$, and $GLCM_{\tau,1}^3$ derived from the 4-order Symmetric Binary Tree (SBT) shown in Fig. 2. The proposed $GLCM_{\tau,s}^m$ encodes the local spatial symmetry relation of the grayscale information distributed over the branch pair $(B_{\tau,s}^{r,m}, B_{\tau,s}^{l,m})$. In particular, when the gray values distributed over the branch $B_{\tau,s}^{r,m}$ and $B_{\tau,s}^{l,m}$ are completely symmetric about the trunk $T_{\tau,s}$, all the non-diagonal elements of $GLCM_{\tau,s}^m$ will be zero.

Many texture features can be measured from the gray-level co-occurrence matrix [28]. Among them, we choose Entropy, Dissimilarity and Autocorrelation which are defined as follows.

$$FE_{\tau,s}^m = \frac{1}{2 \log Q} \left(- \sum_{u=1}^Q \sum_{v=1}^Q GLCM_{\tau,s}^m(u, v) \cdot \log GLCM_{\tau,s}^m(u, v) \right), \quad (6)$$

$$FD_{\tau,s}^m = \sum_{u=1}^Q \sum_{v=1}^Q \frac{|u - v|}{Q - 1} \cdot GLCM_{\tau,s}^m(u, v), \quad (7)$$

and

$$FA_{\tau,s}^m = \sum_{u=1}^Q \sum_{v=1}^Q \frac{u \cdot v}{(Q - 1)^2} \cdot GLCM_{\tau,s}^m(u, v). \quad (8)$$

Then for the contour point $z(\tau)$, we can construct a feature representation by concatenating all the features extracting by the proposed symmetric binary tree $SBT_{\tau,s}^M$ to form a vector

$$\Phi(\tau) = \{FE_{\tau,s}^m, FD_{\tau,s}^m, FA_{\tau,s}^m, \alpha_{\tau,s} | m = 0, \dots, M - 1, s = 1, \dots, S\}, \quad (9)$$

where M is the number of the orders and S is the number of scales for the used $SBT_{\tau,s}^M$. It is worth noting that since the spreading angle $\alpha_{\tau,s}$ of $SBT_{\tau,s}^M$ is good at characterizing the shape property of the object which can be regarded as a complementary information to the feature representation, we also included it in the feature vector $\Phi(\tau)$ which forms a final feature representation of length $(M + 1) \cdot S$. We term it SBT descriptor.

3.3. Characters of SBT Descriptor

In this subsection, we summarize the desirable properties of the proposed SBT descriptor. Although there is a consensus that co-occurrence features are more discriminative than individual occurrence features [16], how to construct effective co-occurrence feature representation adaptive to different tasks of object recognition is still an open issue. As introduced in Section 3.1, the proposed SBT is a symmetric binary tree-like geometric structure that has multiple orders of branch pair $(B_{\tau,\rho}^{r,m}, B_{\tau,\rho}^{l,m})$ and each branch pair is restricted to be symmetric about the same symmetrical axis, i.e. the trunk $T_{\tau,\rho}$. Therefore, the proposed SBT descriptor is a novel co-occurrence feature representation that has large support regions

and can capture strong spatial correlation between texture features which benefits the extraction of more subtle and complex structure of leaf images.

The proposed SBT descriptor depicts the leaf images at multiple scales which make it informative for distinguishing leaf images. In addition, the proposed SBT descriptor is inherently invariant to rotation, scaling and translation and has the local characteristics which make it insensitive to geometric variations of leaf images.

3.4. Matching SBT Descriptors

In this subsection, we use the proposed SBT feature representation $\Phi(\tau)$ to measure the dissimilarity between two leaf images Γ and Δ . We first uniformly sample N points $z_\Gamma(\tau_0), z_\Gamma(\tau_1), \dots, z_\Gamma(\tau_{N-1})$ from the outer contour of leaf image Γ and N points $z_\Delta(\tau_0), z_\Delta(\tau_1), \dots, z_\Delta(\tau_{N-1})$ from the outer contour of leaf image Δ , respectively. Then for each leaf image, we collect SBT descriptor from each contour point to form a set of SBT descriptors. We denote the set $SBT_\Gamma = \{\Phi_\Gamma(\tau_0), \Phi_\Gamma(\tau_1), \dots, \Phi_\Gamma(\tau_{N-1})\}$ for the leaf image Γ and the set $SBT_\Delta = \{\Phi_\Delta(\tau_0), \Phi_\Delta(\tau_1), \dots, \Phi_\Delta(\tau_{N-1})\}$ for the leaf image Δ , respectively.

Hausdorff distance is an efficient and effective metric to measure how far two subsets of a metric space are from each other and has many variants applied for different tasks. We choose the version of the Hausdorff distance introduced in [29] for measure the distance between the set SBT_Γ and the set SBT_Δ , where we use L_1 -norm to calculate the distance between the element $\Phi_\Gamma(\tau_i)$ in SBT_Γ and the element $\Phi_\Delta(\tau_j)$ in SBT_Δ .

3.5. K-NN Based Handcrafted and Deep Features Fusion (KNN-HDFF)

Recently, deep learning techniques have shown great progress in many applications. Considering that deep features learned from the CNN models have the potential to capture the high level features representing the semantics of the data, we are interested in the fusion of handcrafted features with deep features and mining their complementary characteristics for improving the retrieval performance in fine-grained leaf image retrieval. In this section, we propose a novel feature fusion scheme for boosting the retrieval accuracy.

Assume that ψ_q is a query image and $\psi_{d_i}, i = 1, \dots, W$ are the database images, where W is the number of the database images. Let D_{hf} and D_{df} are the pairwise distance measures using handcrafted feature and deep feature, respectively. Then for the query image ψ_q and a database image ψ_{d_i} , there are two distances $D_{hf}(\psi_q, \psi_{d_i})$ and $D_{df}(\psi_q, \psi_{d_i})$ measured by different features between them. Since they are defined in different feature spaces, simply adding them together cannot make them coordinated to contribute to the identification task. To effectively fuse them, we propose a novel algorithm, named K-NN based Handcrafted and Deep Feature Fusion (KNN-HDFF). The algorithm is detailed as follows.

Input: The query image ψ_q and the database images $\psi_{d_i}, i = 1, \dots, W$, the distances measures D_{hf} and D_{df} , and the parameter K ($1 \leq K \leq W$).

Output: The fused distance $\bar{D}_{hd}(\psi_q, \psi_{d_i})$ between ψ_q and $\psi_{d_i}, i = 1, \dots, W$.

- (1) For the query image ψ_q , calculate its distances to all the database images using the measure D_{hf} : $D_{hf}(\psi_q, \psi_{d_j}), j = 1, \dots, W$ and select top K minimum distances from them to form a distance set $SD_{hf}^{(q)}$. Similarly, we build a set $SD_{df}^{(q)}$ of K minimum distances for the measure D_{df} .
- (2) For each database image ψ_{d_i} , conduct following operations:
 - i calculate its distances to the query image ψ_q and all the other database images using the measure D_{hf} : $D_{hf}(\psi_q, \psi_{d_i})$

and $D_{hf}(\psi_{d_i}, \psi_{d_j}), j = 1, \dots, i-1, i+1, \dots, W$ and select top K minimum distances from them to form a distance sets $SD_{hf}^{(i)}$. Similarly, we build a set $SD_{df}^{(i)}$ of K minimum distances for the measure D_{df} .

- ii Calculate the mean value $\bar{D}_h$ over the union set $SD_{hf}^{(q)} \cup SD_{hf}^{(i)}$ and the mean value $\bar{D}_d$ over the union set $SD_{df}^{(q)} \cup SD_{df}^{(i)}$
- iii The fused distance $\bar{D}_{hd}(\psi_q, \psi_{d_i})$ between the query image ψ_q and the database image ψ_{d_i} is calculated as

$$\bar{D}_{hd}(\psi_q, \psi_{d_i}) = D_{hf}(\psi_q, \psi_{d_i}) / \bar{D}_h + D_{df}(\psi_q, \psi_{d_i}) / \bar{D}_d, \quad (10)$$

- (3) Return the fused distances $\bar{D}_{hd}(\psi_q, \psi_{d_i}), i = 1, \dots, W$.

It can be observed From Eq. (10) that the proposed KNN-HDFF algorithm utilizes the neighbour information of the query image ψ_q and the database image ψ_{d_i} , encoded in the variables $\bar{D}_h$ and $\bar{D}_d$, to fuse different distance measures. Using the algorithm KNN-HDFF, for the query image ψ_q , we can obtain the fused distances of it to each database image $\bar{D}_{hd}(\psi_q, \psi_{d_i})$ which are further used for ranking the database images to return the retrieval results.

4. Experimental Results and Discussions

To examine the effectiveness of the proposed Symmetric Binary Tree (SBT) descriptors for addressing the challenging issue of fine-grained leaf image retrieval, we conduct extensive experiments on several fine-grained leaf image datasets and compare it with the state-of-the-arts including handcrafted descriptors and deep learning features using the standard evaluation metrics. In addition, we also perform extensive experiments to validate the effectiveness of the proposed K-NN based Handcrafted and Deep Feature Fusion (KNN-HDFF) method on boosting the performance of fine-grained image retrieval.

4.1. Evaluation protocol

Bulls-eye test is a standard metric that has been frequently used for evaluating the retrieval performance by many methods [30,14,5,4,1]. In this metric, each sample in the database is in turn used as a query to match all the samples in the database. Assume that it has h relevant samples in the database. Among the top 2h matches, count the number of the correct matches to the query as c . The retrieval rate for the query is defined as c/h . The bulls-eye score is defined as the average of the retrieval rates of all the queries. To visualize the retrieval results for observing the behaviours of algorithms, we also apply another commonly used metric, precision-recall curve, for performance evaluation. Assume that there are W samples in the database. Each of them is in turn used as a query $q_i, i = 1, \dots, W$, to retrieve the samples relevant to it from the database. Let $1 \leq j \leq W$ be the number of retrieved samples, c be the number of the samples relevant to the query among the j retrieved samples, h be the number of all the samples relevant to the query in the database. Then the precision and recall for the query q_i under the j samples retrieved are defined as $P_{i,j} = c/j$ and $R_{i,j} = c/h$, respectively. We calculate their average values over all the queries as $\bar{P}_j = \frac{1}{W} \sum_{i=1}^W P_{i,j}$ and $\bar{R}_j = \frac{1}{W} \sum_{i=1}^W R_{i,j}$, respectively. The precision-recall curve is formed by plotting a sequence of points $(\bar{R}_j, \bar{P}_j)$ for $j = 1, \dots, W$.

4.2. Baseline Methods

Six state-of-the-art handcrafted methods including two shape descriptors: MDM [6] and HoGCV [3], two texture descriptors: RLCBP [32] and PRICoLBP [16], a descriptor that combines shape and

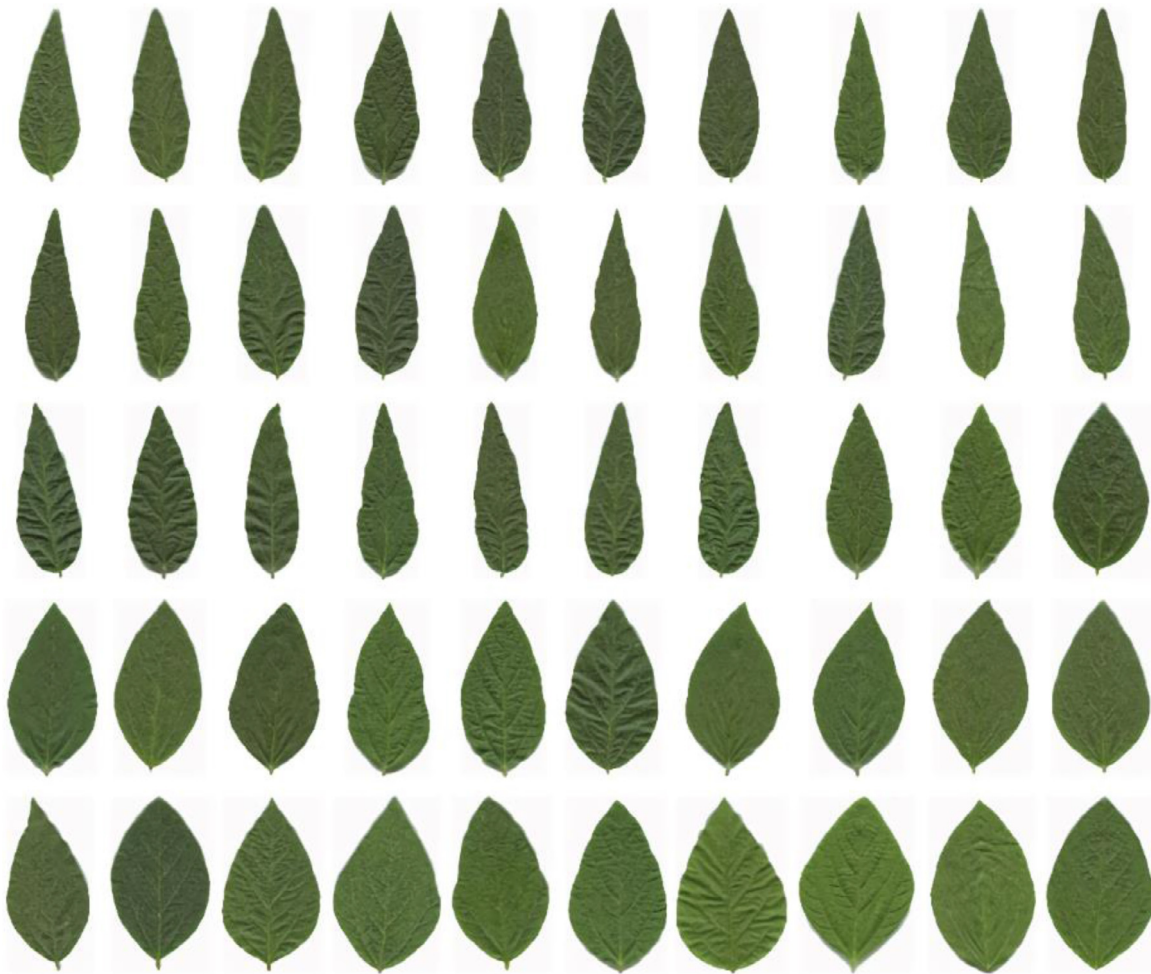


Fig. 6. Parts of leaf images of different cultivars (one example sample for each cultivar) in the Lower Part subset of the SoyCultivar200 dataset [31].

appearance features: MSCM [31], a descriptor that combines shape and texture features: MTD+LBP-HF [27], are used as benchmark algorithms.

Besides handcrafted features, we also compare our method with deep learning features. Two publicly available pre-trained CNN models, ResNet50 [18] and DenseNet121 [33], have been popularly used as generic feature extractors and shown excellent performance in various identification tasks. In this study, we are also interested in their performance on the challenging task of fine-grained leaf image retrieval. In our experiments, the output of the last layer of each CNN model (before the classification layer) is used as the deep feature representation.

Recently, many deep learning features derived from pre-trained or fine-tuned deep networks have particularly been designed for image retrieval and their open source implementation have also been publicly available for facilitating the progress of the research community of image retrieval. We download the source codes of four recently published deep feature representation methods that are designed for image retrieval, including two versions of the Deep-Seated Features Histogram methods, H_{FCLF} and DSFH [36], and two versions of Replicator equation based Selecting and Weighting methods, ReSW and siaMac+ReSW [37]. Among them, H_{FCLF} is derived from the pretrained VGG16 model which is widely used in context of image retrieval, DSFH is obtained by whitening H_{FCLF} with PCA, ReSW is generated from the pretrained VGG16 model and siaMac+ReSW is extracted using the fine-tuned VGG from siaMAC [39]. For all the feature representations, L_1 -norm is used as distance measure.

Then there is a total of six handcrafted descriptors and six deep feature representations used as baselines in our experiments. On the one hand, we will compare the proposed SBT descriptor with them on the benchmark datasets to show its superior performance for fine-grained leaf image retrieval. On the other hand, we will use the proposed feature fusion method KNN-HDFF to fuse each handcrafted descriptor (including our proposed SBT descriptor) with each deep feature representation (which results in a total of $7 \times 6 = 42$ feature fusions) to boost the performance of fine-grained image retrieval. This group of experiments are particularly designed for validating the effectiveness of the proposed method KNN-HDFF for feature fusion and showing the stronger complementarity of the proposed SBT descriptor with deep feature representations.

4.3. Soybean Leaf Dataset

The first benchmark dataset used in our experiments is SoyCultivar200 [31] which is a publicly available leaf image dataset¹ built by collecting leaves from soybean plants of different cultivars that grew in Heilongjiang Province, known as a major soybean production area in China. All the leaves of the SoyCultivar200 belong to 200 cultivars with each having 30 samples that are collected from different parts of soybean plants. For each cultivar, its 10 samples are collected from the upper part, 10 samples from the

¹ Download link: <https://github.com/NUFE-AIAG/FLIR>

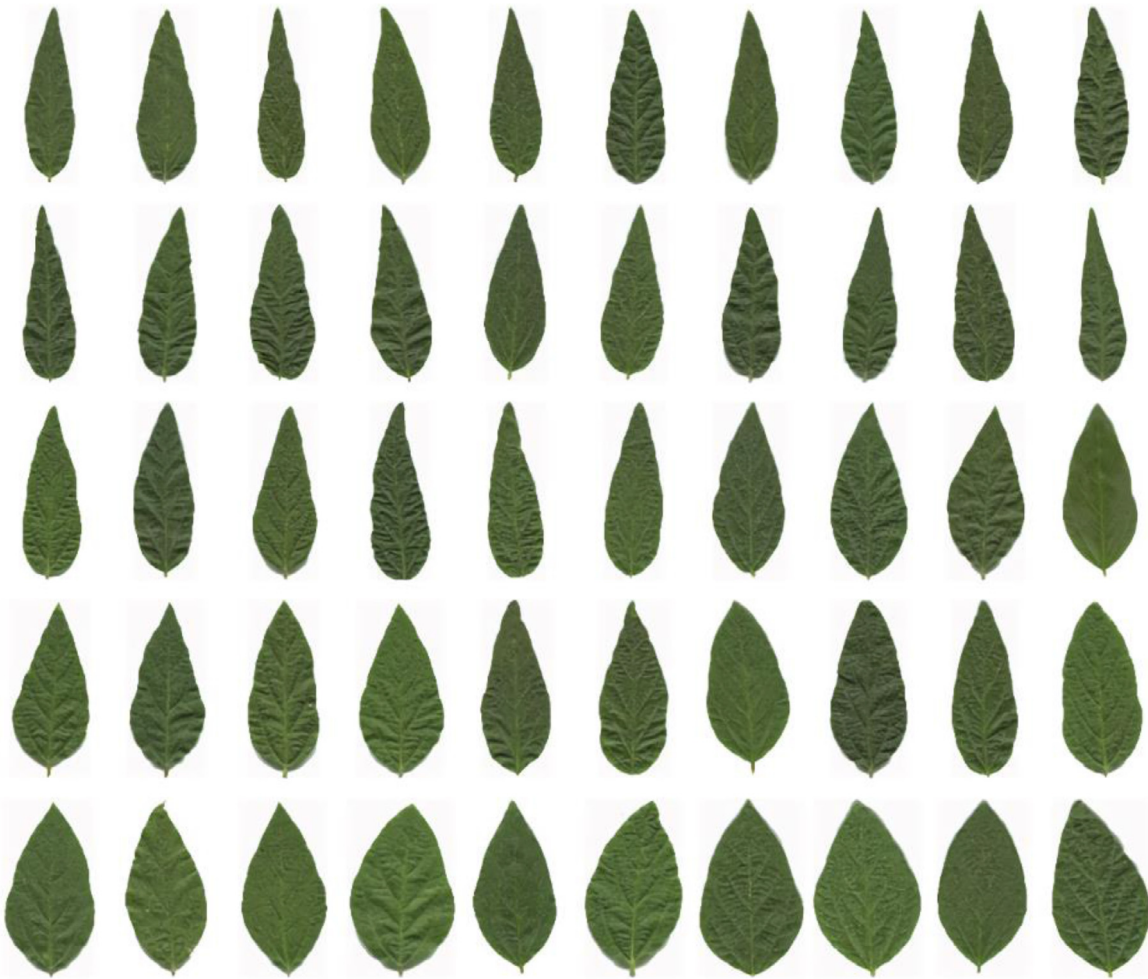


Fig. 7. Parts of leaf images of different cultivars (one example sample for each cultivar) in the Middle Part subset of the SoyCultivar200 dataset [31].

middle part, and 10 samples from the lower part of different soybean plants.

In [31], the SoyCultivar200 are designed for testing the performances of single leaf image pattern matching and joint leaf image pattern matching, respectively. In the former, all the samples from the same part of soybean plants are grouped into the same subset which makes the dataset divided into three subsets, Upper Part, Middle Part and Lower Part. Each subset has $200 \times 10 = 2000$ single leaf patterns. Parts of leaf images of different cultivars from the three subsets are shown in Figs. 6–8, respectively. From them, we can see that many cultivars have high similar leaf image patterns which makes the three subsets become very challenging fine-grained leaf image datasets. While in the later, all the 6000 leaves in the dataset are divided into 2000 groups with each containing three leaves that are from the upper part, middle part and lower part of the same cultivar, respectively to form a joint leaf pattern. Then the dataset has $200 \times 10 = 2000$ joint leaf patterns. The former tests the discriminative power of an algorithm on the use of single leaf image pattern. While the later test the discriminative power of an algorithm on the use of joint leaf image pattern.

The retrieval results of the proposed method and all the 12 benchmark approaches on the three subsets, Upper Part, Middle Part, and Lower Part of the SoyCultivar200 dataset [31], using single leaf image pattern matching, and using joint leaf image pattern matching, are summarized in Table 1. The parameters of the proposed SBT descriptor are empirically set to $M = 4$, $S = 8$ and $N = 256$. It is encouraging to observe that using single leaf im-

age pattern matching, the proposed method achieves the accuracies of 47.57%, 48.40% and 47.92% on the subsets, Upper Part, Middle Part and Lower Part, respectively which are all higher than the benchmark approaches. While using joint leaf image pattern matching, the proposed method achieves an exciting accuracy of 81.69% which is over 0.72% higher than the deep feature representations and is over 5.04% higher than the handcrafted features.

The precision-recall curves of the proposed method and all the benchmark approaches on the three subsets, Upper Part, Middle Part, and Lower Part of the SoyCultivar200 dataset [31], using single leaf image pattern matching, and on the whole dataset using joint leaf image pattern matching, are shown in Fig. 9. As can be seen that the proposed method achieves the best precision-recall curves among all the competing methods for the Lower Part and Joint test cases and is comparable with the deep learning method siaMAC+ReSW and outperforms the other 10 competing methods on the Middle Part and Upper Part test cases.

4.4. Peanut Leaf Dataset

To further examine the effectiveness of the proposed method for fine-grained leaf image retrieval, we collect 500 leaves of 100 peanut cultivars with 5 sample per cultivar to build a peanut leaf image dataset, named PeanCultivar100². All the peanut leaves are randomly collected from the upper part of different plants. An ex-

² Download link: <https://github.com/NUFE-AIAG/ SBT>

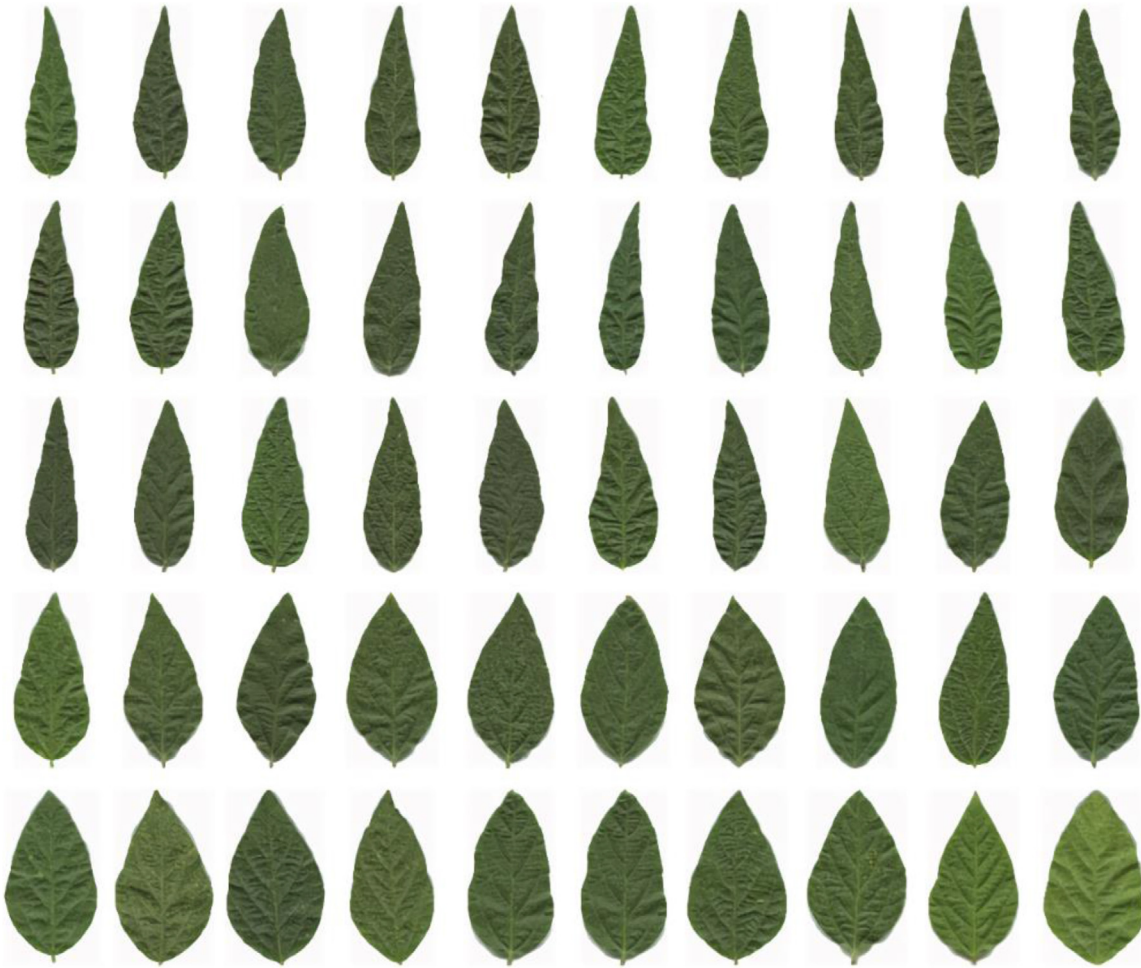


Fig. 8. Parts of leaf images from different cultivars (one example sample for each cultivar) in the Upper Part subset of the SoyCultivar200 dataset [31].

Table 1

The Bull-eye scores (%) of various methods on the SoyCultivar200 leaf dataset [31] by using single leaf image pattern matching and joint leaf image matching.

Algorithm	Upper Part	Middle Part	Lower Part	Joint Matching
MDM [6]	22.94	24.44	25.69	43.26
HoGCV [3]	29.04	31.48	32.48	59.81
RICLBP [32]	31.06	31.50	30.67	51.41
PRICoLBP [16]	35.27	34.86	32.47	56.22
MSCM [31]	34.07	36.62	37.40	64.53
MTD+LBP-HF [27]	40.79	43.13	41.51	76.65
ResNet50 [18]	30.90	34.16	33.02	65.65
DenseNet121 [33]	36.58	38.18	36.95	71.03
H _{FCLF} [36]	38.35	38.5	36.13	66.14
DSFH [36]	41.52	41.82	40.35	69.90
ReSW [37]	45.12	46.33	43.78	80.97
siaMAC+ReSW [37]	47.32	48.35	44.25	78.58
Proposed	47.57	48.40	47.92	81.69

ample sample for each peanut cultivar is shown in Fig. 10 and ten example cultivars and their all the samples are shown in Fig. 11. It can be observed that most of them have very similar leaf image patterns making it become another challenging fine-grained leaf image dataset. The same evaluation metrics, benchmark methods and algorithm parameters of the proposed method as the former groups of experiments are used in this test.

The retrieval rates of the proposed method and the benchmark methods are summarized in Table 2. It can be seen that on the challenging PeanCultivar100 leaf dataset, the proposed achieves a

Bullseye score of 50.12% which is 5.60% higher than the benchmark approaches. Fig. 12 plots the precision-recall curves for all the competing methods. As can be seen that the proposed has the better precision-recall curves than the benchmark methods.

4.5. Fusing Handcrafted and Deep Features

To examine the effectiveness of the proposed KNN-HDFF algorithm on fusing handcrafted and deep features for fine-grained leaf image retrieval, we conduct it to fuse the seven benchmark handcrafted descriptors with the six deep feature representations re-

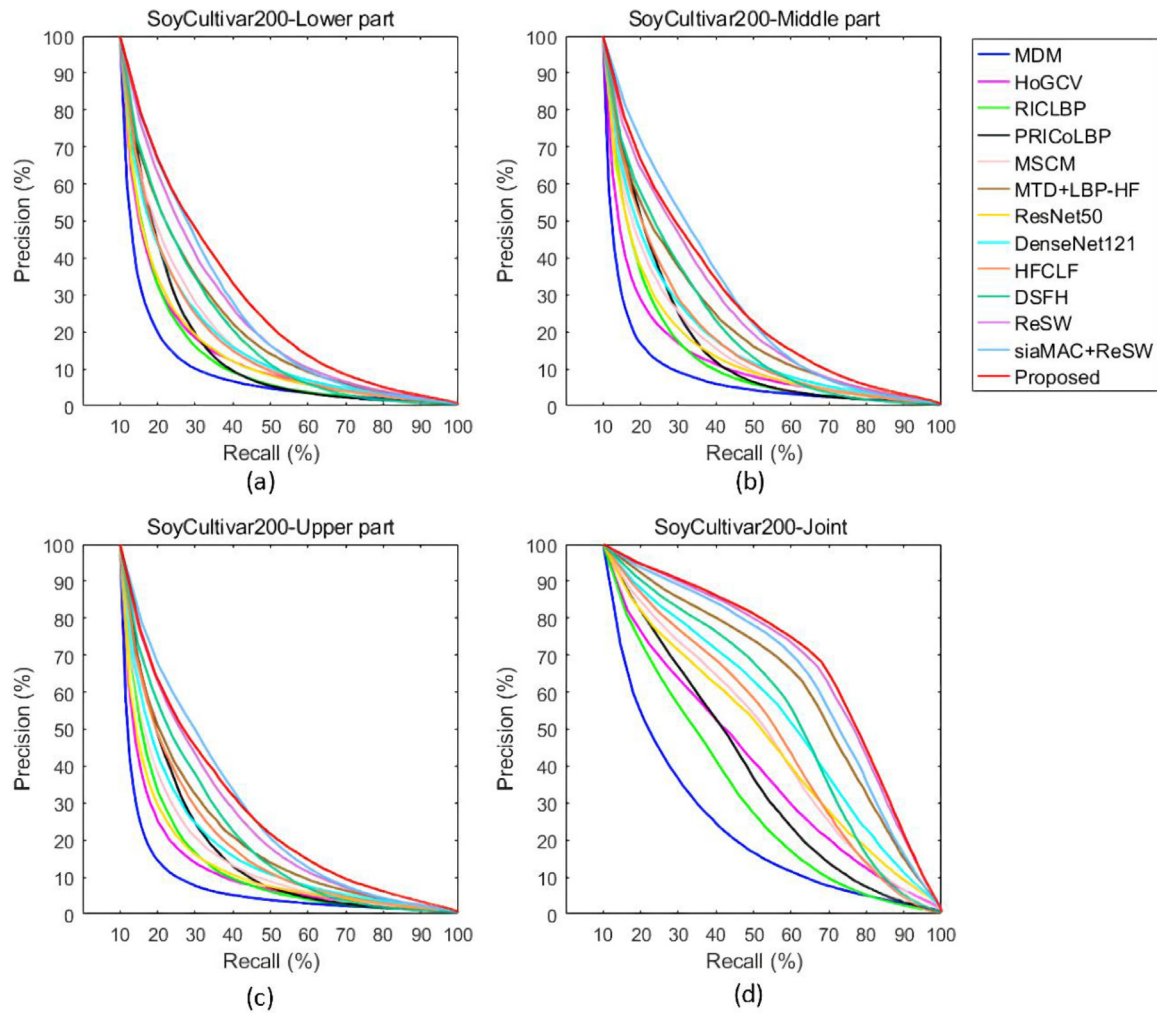


Fig. 9. The precision-recall curves of various methods on the SoyCultivar200 leaf dataset [31]. (a) Lower Part, (b) Middle Part, (c) Upper Part, and (d) Joint matching.

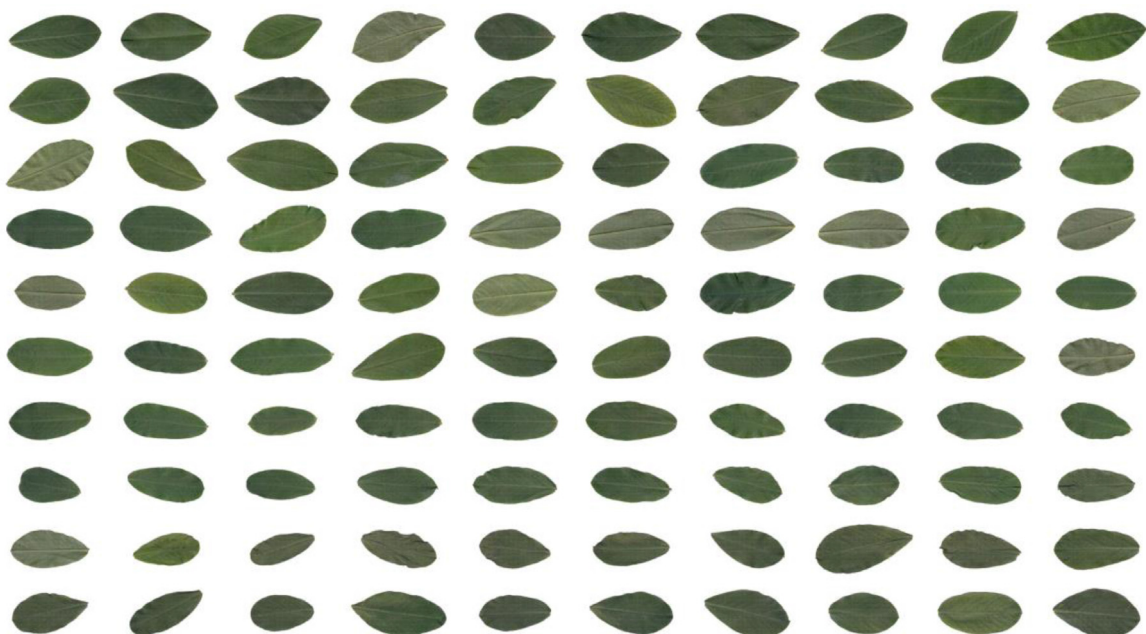


Fig. 10. Example leaf images of different cultivars from the PeanCultivar leaf dataset (one sample for each cultivar).

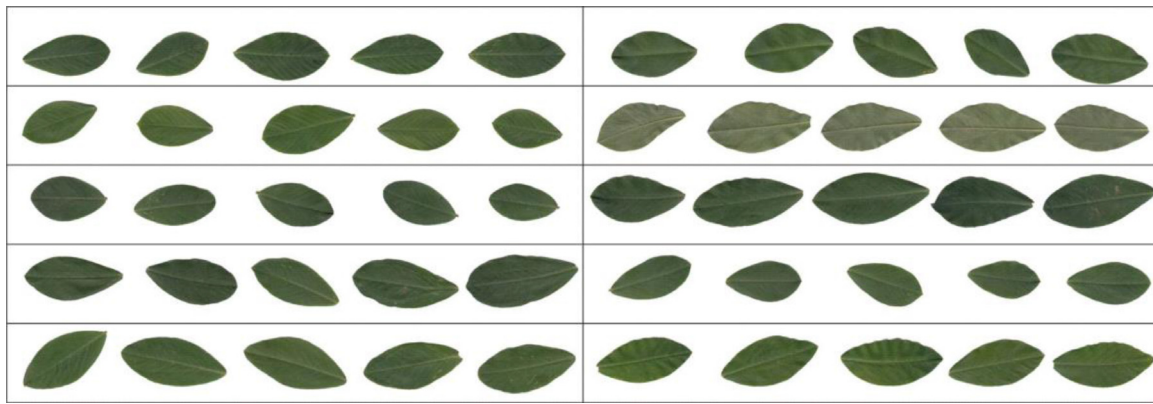


Fig. 11. Ten example cultivars and all the samples of them in the PeanCultivar leaf dataset. Each cell represents a cultivar and all the samples for the cultivar are presented in the cell.

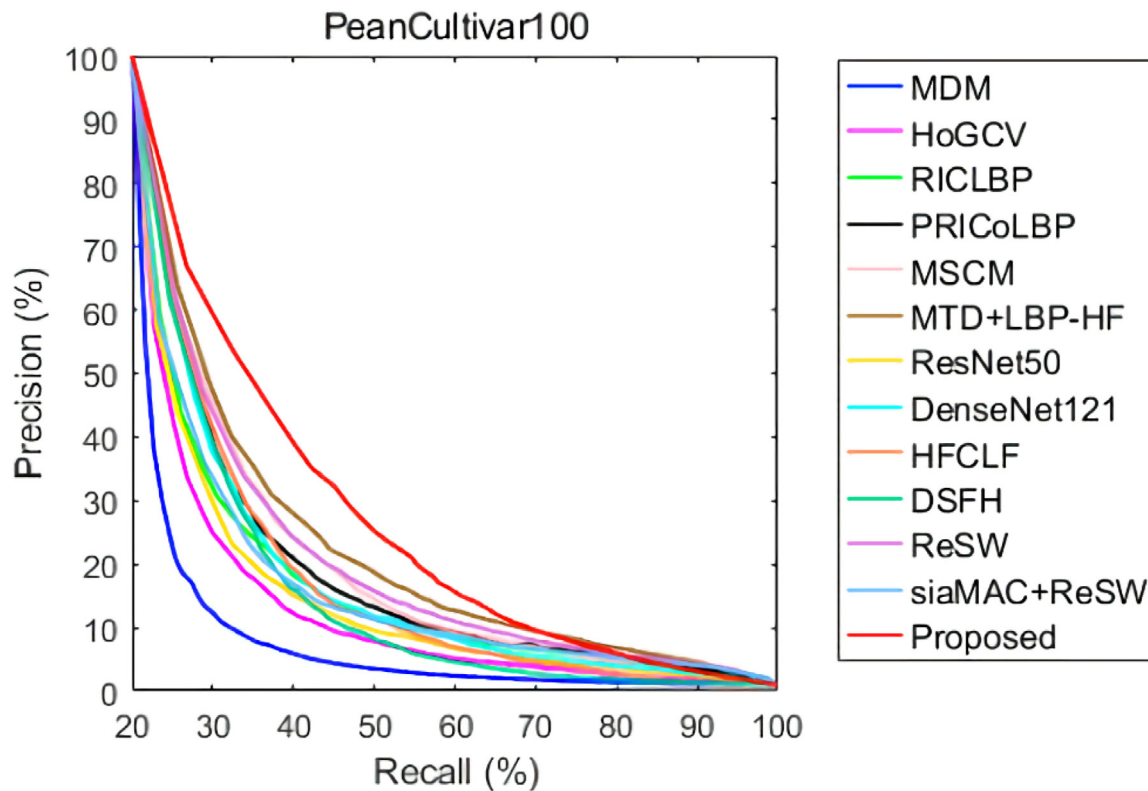


Fig. 12. The precision-recall curves of the proposed method and the benchmark methods on the PeanCultivar100 leaf dataset.

Table 2
The retrieval rates (%) of various methods on PeanCultivar100 leaf dataset.

Algorithm	Bulls-eye score (%)
MDM [6]	28.48
HoGCV [3]	35.16
RICLBP [32]	39.04
PRICoLBP [16]	40.56
MSCM [31]	42.80
MTD+LBP-HF [27]	44.52
ResNet50 [18]	36.44
DenseNet121 [33]	39.24
H _{FCLF} [36]	39.48
DSFH [36]	37.84
ReSW [37]	42.72
siaMAC+ReSW [37]	37.80
Proposed	50.12

spectively. The parameter for the proposed KNN-HDFF algorithm are empirically set to $K = 50$. The retrieval results of various fusions of handcrafted and deep features based on the proposed KNN-HDFF algorithm on the SoyCultivar200 leaf dataset [31] and the PeanCultivar leaf dataset are summarized in Tables 3, 4, 5, 6 and Table 7, respectively.

Comparing the retrieval rates of all the fusions of handcrafted and deep features with their original scores in each Table, we can see that most fusions achieve significant increases of retrieval rates which validate the outstanding performance on feature fusion. For instance, the fusion of MTD+LBP-HF with DSFH make their original retrieval rates have exciting increases of more than 9.8%, 11.95%, 11.43%, 10.26% on the Lower Part, Middle Part, Upper Part, and Joint test cases, respectively. The results of these groups of experiments also demonstrate the stronger complementarity of the proposed SBT descriptor with deep feature representations. The fusion

Table 3

The Bull-eye scores (%) of various fusions of handcrafted and deep features using the proposed KNN-HDFF algorithm on the Lower Part subset of the SoyCultivar200 leaf dataset [31] (The numbers shown in the parentheses are the original scores of the descriptors).

Descriptor	ResNet50 (33.02)	DenseNet121 (36.95)	H _{FCLF} (36.13)	DSFH (40.35)	ReSW (43.78)	siaMAC+ ReSW (44.25)
MDM (25.69)	37.69	39.56	41.79	39.22	44.05	43.48
HoGCV (32.48)	41.97	44.15	46.11	46.23	50.13	49.94
RICLBP (30.67)	38.86	38.65	39.41	35.09	38.16	38.09
PRICoLBP (32.47)	42.21	42.75	42.40	40.03	42.76	42.55
MSCM (37.40)	44.77	48.31	46.72	48.41	50.78	50.75
MTD+ LBP-HF (41.51)	47.52	49.33	49.10	51.31	53.11	53.54
Proposed (47.92)	48.69	51.11	50.15	55.33	56.49	56.08

Table 4

The Bull-eye scores (%) of various fusions of handcrafted and deep features using the proposed KNN-HDFF algorithm on the Middle Part subset of the SoyCultivar200 leaf dataset [31] (The numbers shown in the parentheses are the original scores of the descriptors).

Descriptor	ResNet50 (34.16)	DenseNet121 (38.18)	H _{FCLF} (38.5)	DSFH (41.82)	ReSW (46.33)	siaMAC+ ReSW (48.35)
MDM (24.44)	37.22	39.92	42.88	39.24	44.88	45.72
HoGCV (31.48)	42.60	45.04	48.85	47.08	52.13	52.76
RICLBP (31.50)	41.00	40.82	41.47	36.64	40.10	40.39
PRICoLBP (34.86)	46.58	47.80	47.11	43.51	46.88	46.92
MSCM (36.62)	45.01	47.46	48.94	49.29	52.62	53.07
MTD+ LBP-HF (43.13)	50.16	52.54	53.61	55.08	57.98	58.30
Proposed (48.40)	50.02	53.18	53.64	57.86	60.10	60.01

Table 5

The Bull-eye scores (%) of various fusions of handcrafted and deep features using the proposed KNN-HDFF algorithm on the Upper Part subset of the SoyCultivar200 leaf dataset [31] (The numbers shown in the parentheses are the original scores of the descriptors).

Descriptor	ResNet50 (30.90)	DenseNet121 (36.58)	H _{FCLF} (38.35)	DSFH (41.52)	ReSW (45.12)	siaMAC+ ReSW (47.32)
MDM (22.94)	33.87	37.20	42.17	37.30	43.33	43.54
HoGCV (29.04)	39.09	42.72	48.07	44.69	50.35	50.81
RICLBP (31.06)	40.42	40.57	41.17	36.41	39.68	39.91
PRICoLBP (35.27)	46.91	48.43	48.24	44.04	47.59	48.03
MSCM (34.07)	42.57	46.15	49.03	49.07	52.71	53.45
MTD+ LBP-HF (40.79)	46.71	50.20	51.81	52.95	55.99	56.38
Proposed (47.57)	48.11	52.57	53.93	58.58	60.87	61.44

Table 6

The Bull-eye scores (%) of various fusions of handcrafted and deep features using the proposed KNN-HDFF algorithm under the joint matching on the SoyCultivar200 leaf dataset [31] (The numbers shown in the parentheses are the original scores of the descriptors).

Descriptor	ResNet50 (65.65)	DenseNet121 (71.03)	H _{FCLF} (66.14)	DSFH (69.9)	ReSW (80.97)	siaMAC+ ReSW (78.58)
MDM (43.26)	70.19	72.18	75.38	67.46	78.43	78.94
HoGCV (59.81)	77.00	78.97	81.32	79.01	84.85	84.72
RICLBP (51.41)	67.49	66.81	67.25	59.37	66.46	66.69
PRICoLBP (56.22)	76.59	76.88	76.46	69.45	76.38	76.34
MSCM (64.53)	79.88	81.86	81.49	82.10	86.46	86.67
MTD+ LBP-HF (76.65)	83.81	85.59	85.39	86.91	88.60	88.30
Proposed (81.69)	85.11	86.57	86.04	89.60	90.58	90.01

Table 7

The Bull-eye scores (%) of various fusions of handcrafted and deep features using the proposed KNN-HDFF algorithm on PeanCultivar100 leaf dataset (The numbers shown in the parentheses are the original scores of the descriptors).

Descriptor	ResNet50 (36.44)	DenseNet121 (39.24)	H _{FCLF} (39.48)	DSFH (37.84)	ReSW (42.72)	siaMAC+ ReSW (37.8)
MDM (28.48)	39.56	40.20	41.72	36.12	43.84	40.92
HoGCV (35.16)	45.72	46.44	47.92	43.28	50.20	47.56
RICLBP (39.04)	43.60	44.12	43.20	41.96	43.08	42.24
PRICoLBP (40.56)	46.72	47.56	46.60	45.36	47.60	45.32
MSCM (42.80)	46.76	50.00	49.04	48.96	52.80	50.60
MTD+ LBP-HF(44.52)	48.96	50.48	50.28	50.64	52.16	50.92
Proposed (50.12)	50.64	52.72	51.72	53.36	56.04	53.88

of the SBT descriptor with deep feature representations achieves the best retrieval rates, 56.49% for Lower Part, 60.10% for Middle Part, 61.44% for Upper Part, and 90.58% for Joint on the SoyCultivar200 leaf dataset, and 56.04% for PeanCultivar leaf dataset, among all the fusions of the other handcrafted descriptors with deep feature representations.

5. Conclusion

Identifying leaf images at cultivar level is a very fine-grained pattern recognition problem which requires sufficiently powerful feature representation to capture the subtle difference across different cultivars. This study reports our attempt to retrieval leaf images at cultivar level which has not yet receive considerable attention in computer vision and pattern recognition community. We propose to use a novel geometric configuration, symmetric binary tree (SBT), to produce co-occurrence texture features which has large support regions and can capture strong spatial correlation between texture features. Extensive experiments have been conducted on the soybean cultivar leaf image dataset and the peanut cultivar leaf image dataset. All the experiments consistently indicate that the proposed methods outperform the state-of-the-art methods on fine-grained leaf image retrieval.

It is usually very difficult to greatly improve the accuracy of fine-grained leaf image retrieval by only using handcrafted features or deep features as shown in Table 1 and Table 2. We propose a novel feature fusion method, named K-NN based handcrafted and deep features fusion (KNN-HDFF) that encodes the neighbouring information of distance measure, for boosting the performance of fine-grained leaf image retrieval. Extensive experiments of fusing handcrafted features with deep features have been conducted. The experimental results demonstrate that the proposed method KNN-HDFF can improve the retrieval performance in a large margin. In our future work, we will develop more effective and efficient co-occurrence patterns and feature fusion method for fine-grained leaf image retrieval.

Declaration of competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

References

- [1] B. Wang, Y. Gao, Hierarchical string cuts: a translation, rotation, scale and mirror invariant descriptor for fast shape retrieval, *IEEE Trans. on Image Proc.* 23 (9) (2014) 4101–4111.
- [2] T. Ojala, M. Pietikainen, T. Maenpää, Multiresolution gray-scale and rotation invariant texture classification with local binary patterns, *IEEE Trans. on Patt. Anal. and Mach. Intel.* 24 (7) (2002) 971–987.
- [3] X. Chen, B. Wang, Invariant leaf image recognition with histogram of Gaussian convolution vectors, *Comput. Electron. Agric.* 178 (2020) 105714.
- [4] S. Belongie, J. Malik, J. Puzicha, Shape matching and object recognition using shape contexts, *IEEE Trans. on Patt. Anal. and Mach. Intel.* 24 (4) (2002) 509–522.
- [5] H. Ling, D.W. Jacobs, Shape classification using the inner-distance, *IEEE Trans. on Patt. Anal. and Mach. Intel.* 29 (2) (2007) 286–299.
- [6] R. Hu, W. Jia, H. Ling, D. Huang, Multiscale distance matrix for fast plant leaf recognition, *IEEE Trans. Image Proc.* 21 (11) (2012) 4667–4672.
- [7] B. Wang, D. Brown, Y. Gao, J. La Salle, MARCH: Multiscale-arch-height description for mobile retrieval of leaf images, *Info. Sci.* 302 (2015) 132–148.
- [8] E.T.L. van Bueren, J.R. Myers, Organic Crop Breeding: Integrating Organic Agricultural Approaches and Traditional and Modern Plant Breeding Methods, *Organic Crop Breed.* (2011) 3–13.
- [9] Z. Tang, Y. Su, M.J. Er, F. Qi, L. Zhang, J. Zhou, A local binary pattern based texture descriptors for classification of tea leaves, *Neurocomputing* 168 (2015) 1011–1023.
- [10] Y.G. Nareesh, H.S. Nagendraswamy, Classification of medicinal plants: an approach using modified LBP with symbolic representation, *Neurocomputing* 173 (2016) 1789–1797.
- [11] S.H. Lee, C.S. Chan, S.J. Mayo, P. Remagnino, How deep learning extracts and learns leaf features for plant classification, *Patt. Recog.* 71 (2017) 1–13.
- [12] J.W. Tan, S.W. Chang, S.B.A. Kareem, H.J. Yap, K.T. Yong, Deep learning for plant species classification using leaf vein morphometric, *IEEE/ACM Trans. on Comp. Bio. and Bioinfo.* 17 (1) (2018) 82–90.
- [13] G.L. Grinblat, L.C. Uzal, M.G. Larese, P.M. Granitto, Deep learning for plant identification using vein morphological patterns, *Comp. and Elec. in Agri.* 127 (2016) 418–424.
- [14] C. Xu, J. Liu, X. Tang, 2D shape matching by contour flexibility, *IEEE Trans. on Patt. Anal. and Mach. Intel.* 31 (1) (2008) 180–186.
- [15] J.E. Cavassim, J.C. Bessalho Filho, L.F. Alliprandini, R.A. De Oliveira, E. Daros, E.P. Guerra, AMMI analysis to determine relative maturity groups for the classification of soybean genotypes, *J. of Agronomy* 12 (4) (2013) 168–178.
- [16] X. Qi, R. Xiao, C.G. Li, Y. Qiao, J. Guo, X. Tang, Pairwise rotation invariant co-occurrence local binary pattern, *IEEE Trans. on Patt. Anal. and Mach. Intel.* 36 (11) (2014) 2199–2213.
- [17] M. Minervini, A. Fischbach, H. Scharr, S.A. Tsaftaris, Finely-grained annotated datasets for image-based plant phenotyping, *Patt. Recog. Lett.* 81 (2016) 80–89.
- [18] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [19] J.S. Cope, D. Corney, J.Y. Clark, P. Remagnino, P. Wilkin, Plant species identification using digital morphometrics: A review, *Expert Sys. with App.* 39 (8) (2012) 7562–7573.
- [20] P. Novotný, T. Suk, Leaf recognition of woody species in Central Europe, *Biosys. Eng.* 115 (4) (2013) 444–452.
- [21] X.F. Wang, D.S. Huang, J.X. Du, H. Xu, L. Heutte, Classification of plant leaf images with complicated background, *Appl. Math. and Comp.* 205 (2) (2008) 916–926.
- [22] K. Horaisová, J. Kukal, Leaf classification from binary image via artificial intelligence, *Biosystems Engineering* 142 (2016) 83–100.
- [23] B. Wang, Y. Gao, C. Sun, M. Blumenstein, J. La Salle, Chord bunch walks for recognizing naturally self-overlapped and compound leaves, *IEEE Trans on Image Proc.* 28 (12) (2019) 5963–5976.
- [24] J. Chaki, R. Parekh, S. Bhattacharya, Plant leaf recognition using texture and shape features with neural classifiers, *Patt. Recog. Lett.* 58 (2015) 61–68.
- [25] J. Charters, Z. Wang, Z. Chi, A.C. Tsoi, D.D. Feng, Eagle: a novel descriptor for identifying plant species using leaf lamina vascular features, in: *2014 IEEE international conference on multimedia and expo workshops (ICMEW)*, 2014, pp. 1–6.
- [26] M.G. Larese, R. Namías, R.M. Craviotto, M.R. Arango, C. Gallo, P.M. Granitto, Automatic classification of legumes using leaf vein image features, *Patt. Recog.* 47 (1) (2014) 158–168.
- [27] C. Yang, Plant leaf recognition by integrating shape and texture features, *Patt. Recog.* 112 (2021) 107809.
- [28] W. Gómez, W.C.A. Pereira, A.F.C. Infantosi, Analysis of co-occurrence texture statistics as a function of gray-level quantization for classifying breast ultrasound, *IEEE Trans. on Med. Imaging* 31 (10) (2012) 1889–1899.
- [29] C. Gope, N. Kehtarnavaz, Affine invariant comparison of point-sets using convex hulls and Hausdorff distances, *Patt. Recog.* 40 (1) (2007) 309–320.
- [30] L.J. Latecki, R. Lakämper, T. Eckhardt, Shape descriptors for non-rigid shapes with a single closed contour, in: *2000 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2000, pp. 424–429.
- [31] B. Wang, Y. Gao, X. Yuan, S. Xiong, X. Feng, From species to cultivar: Soybean cultivar recognition using joint leaf image patterns by multiscale sliding chord matching, *Biosys. Eng.* 194 (2020) 99–111.
- [32] R. Nosaka, C.H. Suryanto, K. Fukui, Rotation invariant co-occurrence among adjacent LBPs, in: *Asian Conference on Computer Vision (ACCV)*, 2012, pp. 15–25.
- [33] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *2017 IEEE conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.
- [34] J. Wäldchen, P. Mäder, Plant species identification using computer vision techniques: a systematic literature review, *Archives of Computational Methods in Engineering* 25 (2) (2018) 507–543.
- [35] X. Chen, J. You, H. Tang, B. Wang, Y. Gao, Fine-Grained plant leaf image retrieval using local angle co-occurrence histograms, in: *2021 IEEE International Conference on Image Processing (ICIP)*, 2021, pp. 1599–1603.
- [36] G. Liu, Y. Yang, Deep-seated features histogram: A novel image retrieval method, *Patt. Recog.* 116 (2021) 107926.
- [37] S. Pang, J. Zhu, J. Wang, V. Ordonez, J. Xue, Building discriminative CNN image representations for object retrieval using the replicator equation, *Patt. Recog.* 83 (2018) 150–160.
- [38] H. Tavakoli, P. Alirezazadeh, A. Hedayatipour, A.H. Banijamali Nasib, N. Landwehr, Leaf image-based classification of some common bean cultivars using discriminative convolutional neural networks, *Comp. and Elec. in Agri.* 181 (2021) 105935.
- [39] F. Radenović, G. Tolias, O. Chum, CNN image retrieval learns from BoW: unsupervised fine-tuning with hard examples, in: *European Conference on Computer Vision*, 2016, pp. 3–20.

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